

Interfacing theories of program with theories of evaluation for advancing evaluation practice: Reductionism, systems thinking, and pragmatic synthesis

Huey T. Chen

Department of Public Health, Mercer University, United States

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ABSTRACT

Theories of program and theories of evaluation form the foundation of program evaluation theories. Theories of program reflect assumptions on how to conceptualize an intervention program for evaluation purposes, while theories of evaluation reflect assumptions on how to design useful evaluation. These two types of theories are related, but often discussed separately. This paper attempts to use three theoretical perspectives (reductionism, systems thinking, and pragmatic synthesis) to interface them and discuss the implications for evaluation practice. Reductionism proposes that an intervention program can be broken into crucial components for rigorous analyses; systems thinking view an intervention program as dynamic and complex, requiring a holistic examination. In spite of their contributions, reductionism and systems thinking represent the extreme ends of a theoretical spectrum; many real-world programs, however, may fall in the middle. Pragmatic synthesis is being developed to serve these moderate-complexity programs. These three theoretical perspectives have their own strengths and challenges. Knowledge on these three perspectives and their evaluation implications can provide a better guide for designing fruitful evaluations, improving the quality of evaluation practice, informing potential areas for developing cutting-edge evaluation approaches, and contributing to advancing program evaluation toward a mature applied science.

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1. Introduction

Program evaluation theories are frameworks driving questions, methodologies, and development of new knowledge, based on assumptions believed to be true. They guide the evaluation activities and processes, allow feedback to the program, and inform practice about achievements or missing/weak links preventing successful outcomes. Program evaluation theories have two levels: theories of program and theories of evaluation. The *theories of program* provide assumptions about how to conceptualize a program for the purposes of analysis, understanding, and evaluation (Chen, 1990; 2015) while the *theories of evaluation* provide assumptions about how to design and conduct a sound and useful evaluation (Alkin, 2013; Shadish, Cook, & Leviton, 1991). These two types of theories are related, but are often discussed separately, thus forfeiting opportunities for understanding their relationships and emerging synergies.

This article attempts to contribute toward filling this gap by discussing these two types of theories and their relationships under three theoretical perspectives: reductionism, systems thinking, and pragmatic synthesis. Reductionism and systems thinking have been used as corner stones to build program evaluation as a discipline. However, these two transdisciplinary perspectives, originated from matured sciences, cannot meet all evaluation needs. Pragmatic synthesis, an original, gross-root perspective, is developed to address unmet evaluation needs. An examination of these three perspectives' theories of program and theories of evaluation can better guide designing fruitful evaluations, improve the quality of evaluation practice, inform potential areas for developing cutting-edge evaluation approaches, and contribute to advancing program evaluation toward a mature applied science.

2. Reductionism

Reductionism is the philosophical belief that a system, complex or not, can be broken down into simpler fundamental components which are easier to analyze, understand, or explain (Kemeny &

E-mail address: chen_h@mercer.edu (H.T. Chen).

Oppenheim, 1956; Nagel, 1961). This belief is rooted in the scientific philosophy of parsimony. Reductionism stresses that the aim of science is to identify, isolate, and assess causal relationships between variables. It has been successfully applied in physical and social sciences. A major attraction of reductionism is the promise of a clean-cut solution to a problem that appears too challenging to solve. Reductionism attempts to demonstrate what scientific research is all about, and has been the primary means that medicine and many other sciences, including behavioral sciences and program evaluation, have advanced in the past. Reductionist theories of program and theories of evaluation are discussed below.

2.1. Theories of program

Reductionist theories of program attempt to identify a few crucial components of an intervention program essential for understanding or assessing its performance or effectiveness. These theories are represented by the following two evaluation approaches: *objective-oriented* approach and *goal attainment* model. The objective-oriented approach, originally developed by Ralph Tyler in 1930s (Tyler, 1942, 1950), may be the earliest model in illustrating reductionist theories of program in literature. According to Tyler, the most important component of a program for performance purposes is the component of goals and objectives, because they serve as criteria for judging the merits of the program. However, goals and objectives can be vague and inconsistent, not stated in clear and measurable terms, or can lack consensus. Tyler stressed that evaluators must work with stakeholders to establish goals and objectives, and to determine the extent to which they have been achieved. If they are not, remedial actions need to be taken to correct the problems. The objective-oriented approach is popular in evaluating education programs (Worthen, Sanders, & Fitzpatrick, 2004).

The goal-attainment model (Morris & Fitz-Gibbon, 1978; Schulberg & Baker, 1968), closely related to the objective-oriented approach, makes explicit that goals (measurable goals) need to be examined in conjunction with an intervention for understanding and assessing its effectiveness. Effectiveness is determined by whether an intervention affects its prescribed outcomes. The goal-attainment model has been used as a foundation for reductionist theories of evaluation as discussed in the next section.

2.2. Theories of evaluation

Reductionist theories of evaluation seek to provide credible evidence for a causal relationship between an intervention and outcome beyond the influence from other factors. During the 1960s, evaluators searched for frameworks and methods for performing precise, causal analyses of relationships between interventions and outcomes. Suchman (1967) introduced the Campbellian validity typology (Campbell & Stanley, 1963; Cook & Campbell, 1979; Shadish, Cook, & Campbell, 2002) to the evaluation community for this purpose. He argued that, although the typology was developed for experimental research, it was highly applicable to program evaluation as well, precisely because it allowed for rigorous analysis of causal relationships between the intervention and the outcomes. Suchman's view was well received and supported in the evaluation community. The advocacy of applying Campbellian validity model for guiding evaluation is called the *experimentation evaluation approach*. The approach argues that evaluators should maximize internal validity (strong evidence of an intervention's effectiveness) by eliminating all threats to internal validity (confounded factors) in evaluations. The majority of evaluators are trained to apply experimental and quasi-experimental methods to eliminate threats to internal validity in

evaluation practice. With its popularity, the reductionist theory of evaluation (the experimentation evaluation approach) has become the dominant tradition for outcome or impact evaluation as discussed in evaluation text books (e.g., Mertens & Wilson, 2012; Rossi, Lipsey, & Freeman, 2004; Wholey, Hatry, & Newcomer, 2010).

2.3. Advantages and challenges in applying reductionism to program evaluation

2.3.1. Advantages

The following are major benefits for applying reductionism to evaluation:

- Assuring rigor of evaluation by eliminating the influence of confounding factors.

Reductionism provides conventional standards and methods for achieving rigor in evaluation. By following this theoretical perspective, evaluation can ensure strong evidence to show that the intervention, rather than other factors, influences the outcomes. Rigor is the most attractive advantage of reductionism. With rigor, evaluators could claim evidence provided by an evaluation as credible and trustworthy.

- Enhancing evaluation's scientific recognition and reputation.

Reductionism has driven scientific advancement across different disciplines for many decades. Evaluations following this tradition gain wide scientific recognition and reputation.

- Serving as a basis for the evidence-based intervention movement.

Reductionism has a strong influence on the current evidence-based intervention movement. Researchers believe rigor as the criterion that matters when evidence is of concern. Many of them argue that only interventions evaluated via randomized control trials (RCTs) are designated as *evidence-based* interventions (Nutbeam, 1999; Speller, Learmonth, & Harrison, 1997).

2.3.2. Challenges

In spite of the merits mentioned above, lessons learned from practice indicate the following challenges that reductionism needs to overcome in the context of program evaluation:

- Ignoring practical or service issues of high interest to stakeholders.

Commonly noted stakeholders' concerns relate to practical issues, such as whether the intervention would be attractive to clients in the real-world, whether the intervention would be suitable for regular implementers to deliver, and whether a typical community organization would be capable to manage the intervention. From reductionism's standpoint, these practical issues may appear trivial, but to stakeholders they could be critical. Yet, in evaluations following reductionism these issues are often not addressed.

- Extrapolating efficacy from controlled settings to real-world setting effectiveness.

Reductionism's theories of evaluation make a strong assumption, that if an intervention is *efficacious* in a controlled setting then it is very likely to be *effective* in the real world. This assumption may be true for exact sciences (e.g. biomedical research) but not necessarily so in social betterment or health promotion programs,

which are subject to influence from contextual factors (Chen, 2015). As a result, an application of reductionism may promote or disseminate an efficacious intervention, which may not be effective in the real world.

- Neglecting complexity.

Advocates of systems thinking – discussed in the next section, argue that a system has holistic properties, which may be overlooked by reducing a program to its components. In other words, the principles and properties that are responsible for the collective order of a system cannot be understood merely by examining its components. Reductionism ignores the complexity of a system, and could provide unrealistic or incorrect solutions for a complicated problem.

3. Systems thinking

Systems thinking offers a view at the other end of the spectrum from reductionism: it indicates that a system cannot be understood by reducing it to smaller parts, but by embracing the complexity that emerges from interconnecting parts. The perspective is often introduced with the phrase: *The whole is more than the sum of its parts*. Systems thinking is very broad and of long standing, being developed as a linkage of multiple disciplines including biology, psychology, physics, computer science, and management (Midgley, 2003; Peters, 2014). Systems thinking is rooted in various systems theories including general systems theory, chaos theory, cybernetics theory, learning organization theory, path independency theory, and so on. These diverse disciplines provide a basis for arguing that systems thinking can be widely applied to different phenomena, including to view and analyze an intervention program as a complex system (Cabrera, Colosi, & Lobdell, 2008).

3.1. Theories of program

Contrasting to reductionism that breaks down a system into simpler parts, systems thinking's theories of program propose an understanding of a system by holistically examining the linkages and interactions between the elements that compose the entire system. Its theories of program are represented by the following fundamental ideas (Mingers & White, 2010: 1148):

- “Viewing the situation holistically, as opposed to reductionistically, as a set of diverse interacting elements within an environment.
- Recognizing that the relationships or interactions between elements are more important than the elements themselves in determining the behavior of the system.
- Recognizing a hierarchy of levels of the systems and the consequent ideas of properties emerging at different levels, and mutual causality, both within and between levels.
- Accepting, especially in social systems, that people will act in accordance with differing purposes or rationalities.”

Since social or public health problems are complex, systems thinking argue interventions often need multi-sectorial responses that address a variety of levels including social determinants. Interventions centered on systems thinking ideas are called complex interventions (Shiell, Hawe, & Gold, 2008), systemic interventions (Midgley, 2006), or system-level interventions (Savigny, Adam, & Organization, 2009). For convenience, this article refers to them as complex interventions. There has been a growing interest in using systems thinking as a framework for designing complex interventions to deal with health care

problems. For example, the Institute of Medicine (IOM) published a report, entitled *Crossing the Quality Chasm* (Institute of Medicine, 2001), which endorsed the idea of using systems thinking to propose principles for redesigning the health care system. The IOM report recognized the health care system as a complex and adaptive system that solicits attention to multilevel influences and system change strategies. Additionally, systems thinking had been used to design large scale public health programs to address problems in areas such as tobacco control (Best, Clark, Leischow, & Trochim, 2007) and tuberculosis care (Menabde, 2009).

One of popular tools for describing and explaining complex relationships within an intervention program is the causal loop diagram (CLD). CLD uses feedback loops to indicate the existence and direction of a causal relationship between variables (the plus/minus sign indicates that two variables change in the same/opposite direction). For example, in the UK government's Foresight Programme (Butland et al., 2007; Savigny & Adam, 2009) applied systems thinking to understand the biological and social complexity of obesity. Experts from multiple disciplines were invited to provide advice and insight on obesity and policies needed to address the problem, by using CLD. The results indicated that obesity is caused by a complex system of determinants. No single determinant was found to dominantly influence obesity prevalence.

3.2. Theories of evaluation

Systems thinking's theories of evaluation are examining the linkages and interactions between the components of the entire system and are assessing how they affect intended and unintended outcomes. Contrasting to its theories of program as being highly innovative and progressive, systems thinking' theories of evaluation, however, are much more conservative as indicated in the literature; they focus mainly on how to apply the traditional approaches and methods for evaluating complex interventions. Even so, advocates have major disagreements about which existing approaches and methods should be applied for such kind of evaluation. On one hand, some advocate applying the experimentation evaluation approach proposed by reductionism to evaluate complex interventions. For example, in 2000, the Medical Research Council published a highly influencing framework (Medical Research Council, 2000) that proposed the application of RCTs for complex interventions. In the evaluation section, the report argued RCTs are the most robust methods for preventing biases when evaluating the efficacy of complex interventions. If individually randomized trials may not be feasible, cluster randomized trials or randomized incomplete block designs are recommended. If experimental methods are not feasible, quasi-experiments or observational designs may be considered. The position was echoed in the revised and updated version of the guidance (Craig et al., 2008) as well as other discussions (Campbell et al., 2000). It becomes apparent that these advocates believe the rigor of the experimentation evaluation approach to be highly desirable for evaluations of complex interventions.

On the other hand, others argue that reductionism's RCTs and linear causal analysis are inadequate methods for analyzing complex problems or interventions. They favor the use of natural experiments or case methods instead of the above suggested (Best & Holmes, 2010).

3.3. Advantages and challenges of applying systems thinking to evaluation

3.3.1. Advantages

- Providing a more complete explanation of how an intervention program works.

An intervention in a system is not an isolated piece, but it connects with other components. By examining interactions among multiple components, systems thinking provides insight on how interactions work and how they affect outcomes.

- Magnifying lens for investigating synergies and emergent behaviors.

By looking at interactions and taking a holistic view, systems thinking may be able to detect synergies or emergent behaviors that are difficult to find by investigating individual components. Unless evaluators take a more holistic view, an evaluation focused mainly on the relationship between the intervention and the outcome can potentially overlook synergies or unintended outcomes.

- Facilitating multidisciplinary collaborations to solve a problem.

Obesity, pandemic diseases, and global warming are examples of complex problems with multiple determinants. As a consequence they require input from multidisciplinary or *trans*-disciplinary collaborations. Systems thinking provides the framework for such collaborations.

3.3.2. Challenges

- Information overload.

Due to multiple origins, systems thinking provides an enormous amount of topics, concepts, models, approaches, and theories from various disciplines; these breadths and complexities can perplex and even discourage evaluators. The followings concepts and models are barely illustrating the tip of the iceberg of the vast abundance of this information: chaos, cybernetics, agent-based modeling, complex adoptive system, scenario planning, stow-flow structure, path dependency, etc. The list goes on and on. In order to allow a wider application of systems thinking, fundamental work in screening and prioritizing these concepts, models, approaches, and theories for evaluation purposes, including providing concrete examples for demonstrating their applicability to evaluation, have to be done first. Despite the contributions made by [Cabrera et al. \(2008\)](#), [Patton \(2011\)](#), and [Williams and Imam \(2007\)](#) in introducing systems thinking to the evaluation community, these tasks have yet to be accomplished to a larger extent.

- Difficulties in data analysis, reporting of findings, and communication of results.

Systems thinking has proposed many concepts such as feedbacks, interactions, nonlinear relationships, multi-level influences, multiplier effects, and so on. As discussed in the previous section, these concepts are essential for its theories of program. However, it is still unclear how to translate these concepts in systems thinking theories of evaluation. Using the Foresight Programme of the UK Government Office ([Butland et al., 2007](#); [Savigny & Adam, 2009](#)) as an example, the system map of the program contains 108 variables. The relationships between variables are illustrated with more than 300 lines indicating positive and negative influences. The connections resulted from feedback loops range from 2 variables to as many as 16 variables. It is unclear how to meaningfully analyze the large amount of causal loops and draw evaluation conclusions about the effectiveness of the program by using exiting statistical techniques, and how to summarize and communicate the results.

- Rely on reductionism's theories of evaluation

Systems thinking seems to oppose to reductionism. On one hand, advocates of systems thinking severely criticize reductionist theories of program. On the other hand, however, many of them embrace reductionist theories of evaluation for evaluating complex interventions. This contradictory position and its implications have not been formally addressed and discussed, neither can they be ignored for further promoting systems thinking in evaluation.

4. Pragmatic synthesis

In spite of their contributions, reductionism and systems thinking represent the extreme ends of a theoretical spectrum; many real-world programs may actually fall in the middle. Evaluators who work on these moderate-complexity programs have no choice but to select one of the two perspectives for guiding their work, often insufficiently or inappropriately addressing the stakeholders' interest and need. It becomes evident that an alternative theoretical perspective must be developed to serve those programs. The author proposes *pragmatic synthesis* as a new perspective to fill this gap. Pragmatic synthesis is an original, grass-root theoretical perspective grew out of the following sources: 1). practice evaluation in communities and feedback from stakeholders, 2) theory-driven evaluation literature, and 3). Contributions made by evaluation theorists including Campbell, Cronbach, and Rossi. Pragmatic synthesis recognizes that the development and implementation of an intervention program in a community setting are challenging, but program planners usually find ways to make dynamic situations more predictable and manageable. Pragmatic synthesis views moderate-complexity programs are characterized as organized efforts to coordinate activities and managing contextual factors to support an intervention for activating its transformation process to bring about desirable outcomes. Components in the system are interconnected, but not all of them directly and actively interact with the intervention to generate effects. Such a system could be understood by modeling major components, connections and interactions of major components. Pragmatic synthesis' view on a system is more complicated than reductionism, but of lesser complexity than systems thinking. Because moderate-complexity programs are different from the programs conceptualized by reductionism, pragmatic synthesis argues evaluation approaches proposed by reductionism may be inappropriate or not fruitful for evaluating moderate-complexity programs. Alternative evaluation theory and methodology must be developed. Pragmatic synthesis' theories of program and theories of evaluation are discussed as follows:

4.1. Theories of program

Pragmatic synthesis' theories of program grew out of two simultaneous movements intending to expand reductionism's theories of program.

1.) Logic models/logframe

Government agencies and private foundations recognized that reductionism's theories of program were too simplistic for program planning and development purposes thus began to promote logic models and logframe as alternatives in 1980s. The Centers for Disease Control and prevention ([CDC, 1999](#)) and private foundations, including the [United Way of America \(1996\)](#) and [Kellogg Foundation \(2004\)](#), promote *logic models* as a tool for guiding program planning and development. Logic models propose to describe an intervention program using the following set of components: inputs, activities, outputs, and outcomes. The relationship among the components is represented by a chain of

“if . . . then . . .” statements. Similarly, the *logframe* (or so called logical framework approach), developed by the U.S. Agency for International Development (USAID), identified the following set of components for planning or developing a program: activities, outputs, purposes, and goals (USAID, 2002). This approach clarified what *activities* should take place, to produce the *outputs* required for accomplishing the *goals* of the project, such that they serve its *purpose*. Each component has narrative description, indicators, information availability, and external influences. Because funding organizations require grantees to apply either logic models or logframe in their programs, these tools/models have been widely used for describing a program.

2) Program theory/theory of change

Also started in 1980s, evaluators began to recognize that the scope of the goal attainment model was too narrow to conceptualize a program for understanding and evaluating an intervention program. The proposal of program theory or theory of change (meaning mechanisms mediating relationships between an intervention and outcome), and its applicability in conceptualization of a program, attempted to address this need (Chen & Rossi, 1980; Chen, 1990; M. Coryn, Noakes, Westine, & Scholer, 2011; Donaldson, 2007; Funnell & Rogers, 2011; Nkwake, 2012; Weiss, 1998). Similar efforts include *realistic evaluation* (Pawson & Tilley, 2007) and *intervention mapping* (Bartholomew, Parcel, & Kok, 1998).

Program theory/theory of change and logic model/logframe are related, but they emphasize different portions of a program: Logic models/logframe accentuate addressing action steps, while program theory/theory of change emphasizes the elaboration of change processes. Pragmatic synthesis' theories of program contain both features and are best represented by the action model/change model schema (Chen, 2005, 2015). The schema conceptualizes an intervention program as a systematic configuration of stakeholders' prescriptive assumptions (how to do it) and descriptive assumptions (why it works) underlying programs, whether explicit or implicit. The descriptive assumptions (*change model*) explain the causal processes expected to happen in order to attain program goals. A change model describing the causal process generated by the program consists of the following components: intervention, determinants, and outcomes. The prescriptive assumptions (*action model*) explain what actions must be taken in a program in order to activate the change model to produce desirable outcome. The action model describes a systematic plan for arranging staff, resources, settings, support organizations, and environmental conditions needed to reach a target group and deliver intervention services. It contains the following components and assumptions:

- Implementing organizations: They must have capacities for organizing and coordinating activities to support the intervention.
- Implementers: They must be competent and committed to provide services.
- Associated organizations and partners: They must be willing to collaborate with the program.
- Ecological context: Community norms, culture, other environmental factors are supportive or at least not against the program.
- Intervention and service delivery protocols: The program must have protocols on exact content and activities for bringing about desirable change, and models of service delivery.
- Target population: The program must have a clearly defined target population and strategies to identify, recruit, and motivate them to participate in the program.

It is important to note that the intervention is just one of the components in the action model for bringing out the desirable changes. For a program to be effective, its action model must be sound and its change model plausible; its implementation is then also likely to be seamless. The components of the schema and their relationships are illustrated in Fig. 1.

Fig. 1 indicates that the action model must be implemented appropriately to activate and support the “transformation” process in the change model. The components in the action model may have a moderating effect on the relationships between the intervention, determinants, and outcomes. Furthermore, feedback happens between the models and the components for adjustments. Interplay between models and components may generate emergent phenomena which affect outcomes.

Pragmatic synthesis' theories of program, represented by the action model/change model schema, provide a possible reconciliation of differences between reductionism and systems thinking on conceptualizing a program. On one hand, it addresses reductionism's potential weakness of over-simplicity. On the other hand, it avoids systems thinking's potential problem of information overload.

4.2. Theories of evaluation

Pragmatic synthesis' theories of evaluation are to assess the transformation resulted from the implementation of the action model, bearing the influence of the change model, and the interplay between them. These evaluations require utilizing both traditional and innovative methodologies. Traditional methodologies used are: qualitative methods (used to help stakeholders clarify their view on the schema), quantitative methods (to assess changes), and mixed methods (to triangulate evidence). Innovative methodologies developed from community experiences are applied to address issues emerged from interconnections between the action model and change model (Chen, 2010, 2015; Chen & Garbe, 2011; Chen & Turner, 2012; Chen et al., 2014). The rest of this section is to briefly introduce innovative views, concepts and approaches offered by pragmatic synthesis:

- Going beyond the doctrine of ruling out all threats to internal validity in evaluation

As discussed in the section of reductionism, the experimentation evaluation approach faithfully follows the Campbellian validity typology. It requires evaluation to rule out all threats to internal validity, so that credible evidence can be obtained outside of confounding effects. According to this doctrine, the more threats to internal validity are ruled out, the better the evaluation is. From an analytical standpoint, the doctrine has merit; pragmatic synthesis, however, provides an alternative view on the issue by arguing that this approach should be considered as an option rather than a universal principle. Stakeholders' interests in specific intervention effects, may at times require placing less emphasis on the rigor of the evaluation methods, thus influencing whether all threats to internal validity be ruled out or not. There are two types of intervention effects: pure independent effects and joint effects. If stakeholders are interested in *pure independent effects*, then it is highly justifiable to rule out all threats to internal validity. However, if the purpose is to assess *joint effects*, there is no need, even inappropriate, to blindly rule out all threats to internal validity.

Pure independent effects are the effects generated by the intervention alone. They are obtained through manipulating and controlling research settings as in RCTs, when establishing a causal relationship is important to researchers. However, pure independent effects are not likely to be found in the real world.

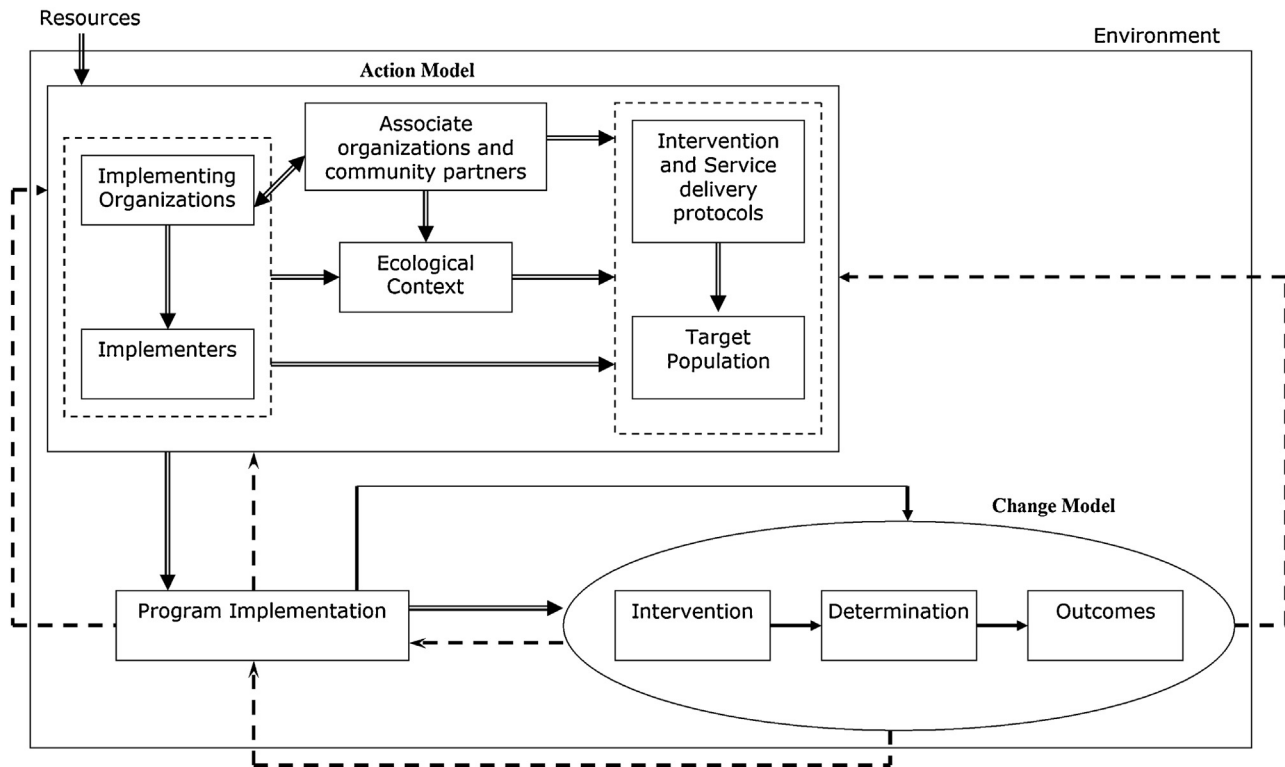


Fig. 1. The action model/change model schema.

Stakeholders may not find them critical to the success of their practice, although they may recognize the merit of the methodology. Typically, stakeholders deliberately take advantage of real-world factors (which could threaten internal validity) and include them into their practice, to help clients. For example, the experimentation evaluation approach views *selection* for participation in an intervention as a validity threat, which must be ruled out. However, community practitioners view selection as an integral part of their practice: Practitioners believe that clients must at least have some interests to join a program, or wish to make changes in their lives in order to benefit from services (Chen, 2015). Accordingly, stakeholders are more interested in seeing the effects created by both, intervention and selection, in the real world, rather than by intervention alone.

In order to better reflect stakeholders' interest, pragmatic synthesis introduces two new concepts: *adjuvants* and *joint effects* (Chen, 2015). *Adjuvants* are factors which can enable an intervention to work, or enhance its effectiveness. Many adjuvants used by stakeholders fall in the category of threats to internal validity. *Joint effects* are emerging from the *interaction* between the intervention and adjuvants. These are effects actually observed in the real world. The new concepts of *adjuvants* and *joint effects* parallels the stakeholders' understanding and experience on how an intervention works, making it easier for evaluators to incorporate them in reports.

The above discussions imply that pure independent effects and joint effects have their own importance for understanding all aspects of the intervention (Chen, 2015). Evaluations should avoid the myth that pure independent effects are superior to joint effects or vice versa. The implications of each type of effect should be discussed with the stakeholders, and those which best serve the stakeholders' needs should be selected. If stakeholders are interested in both and there are sufficient funds to accommodate that need, pure independent effects and joint effects should be included in evaluation.

- Offering the holistic effectuality evaluation approach for assessing real-world effects.

Since the experimentation evaluation approach is assessing pure independent effects, an alternative approach is needed for joint effects/real-world effects. Pragmatic synthesis proposes the *holistic effectuality evaluation approach* for that purpose (Chen, 2015) including methodologies and a checklist of items for ranking real-world evaluation quality. One of the items requires evaluators to apply qualitative methods to identify adjuvants and examine their interconnections with an intervention in producing joint effects.

- Expanding on the scope of credible evidence.

Currently, the scope of credible evidence mainly focuses on issues of *efficacy* or *effectiveness*. For example, evidence-based interventions are evaluated by RCTs. Although this method may serve well for research purposes, it does not capture the length of stakeholders' views and interests. Efficacy or effectiveness is just one piece of evidence for solving the puzzle. Stakeholders are also interested in knowing whether an intervention is capable of surviving or thriving in a community (Chen, 2010). To serve this need, pragmatic synthesis proposes the concept of *viability*, another phenomenon emerging from the interplay between the action model and change model. More specifically, viability refers to the extent to which an intervention is suitable, affordable, practical, helpful, and evaluable. Viability evaluation proposes to assess viability of an intervention program by asking questions like: Can regular practitioners—rather than research staff—routinely implement an intervention program adequately? Is the intervention program suitable for coordination or management by a service delivery organization (such as a community clinic) or a community-based organization? Is the intervention program affordable? Can it recruit participants without offering incentives?

Does the intervention have a clear rationale for its structure and linkages with expected outcomes? Do regular clients and other stakeholders regard the intervention as helpful in alleviating clients' problems or enhancing their well-being? To better address stakeholders' interest, pragmatic synthesis proposes to expand the scope of credible evidence from effectiveness alone to a combination of effectiveness and viability.

- Proposing a realistic version of external validity for evaluation: Transferable cogency.

In addition to internal validity, the experimentation evaluation approach also brings evaluators' attention to the concept of *external validity* as proposed by the Campbellian validity model (Campbell & Stanley, 1963; Cook & Campbell, 1979; Shadish et al., 2002). External validity (Campbell & Stanley, 1963) asks the following question: "To what populations, settings, treatment variables, and measurement variables can this effect be generalized?" In spite of its utility and popularity, this definition of external validity may be more appropriate for academic research than it is for program evaluation. Academic research has great interest in testing a proposition's generalization potential across different populations, settings, treatment variables, and measurements. Campbell and Stanley's definition of external validity fits such academic interest very well. In evaluation, however, such open-ended quest for law-like propositions is extremely difficult or impossible to achieve—evaluation is usually constrained by the resources, timeline, and purposes imposed by funding agencies. Thus, when external validity is conceptualized as an endless quest, the concept itself becomes a barrier that hinders evaluators from pursuing external validity issues. Perhaps, this is one of the major reasons why progress in addressing issues of generalizability has been rather slow in program evaluation.

To address the stagnation, pragmatic synthesis proposes to revise the concept of external validity for fitting evaluation needs. The revised concept is called *transferable cogency*, defined as the extent to which an intervention's effectiveness and viability are transferable from research to real-world setting, or from one real-world setting to another (Chen, 2015). Transferable cogency can be achieved via *targeted generalization* (a particular target population and setting to which stakeholders want to generalize the evaluation results), or *exhibited generalization* (the extent to which an evaluation provides sufficient information about contextual factors for an intervention to be effective in real world applications). Potential users can adapt the information on the effectiveness of the intervention, together with the contextual factors. Users can thereby assess the generalization potential as applicable to their own populations and settings, and decide whether to apply the intervention in their communities. The application of exhibited generalization is illustrated by an evaluation of a carbon monoxide ordinance (Chen et al., 2014).

- Clarifying the myth of inferiority of stakeholder theory-based interventions.

Stakeholder theory-based interventions originate from stakeholders' experience and observation in working with clients and partners in a community. These interventions are viewed by many researchers and evaluators as inferior to formal theory-based interventions that follow the well-travelled path of social/behavioral science theories developed by academia. Pragmatic synthesis recognizes stakeholder theory's weaknesses, but also its merits. Compared to formal theory-based interventions, they tend to be more viable, more likely to get community support, and more likely to be adequately implemented by community-based organizations (Chen & Turner, 2012). Pragmatic synthesis argues

that it is a mistake for researchers and evaluators to only focus on formal theory-based intervention and ignore stakeholders' theory-based interventions. By doing so, they forgo opportunities to learn from communities and minimize the extent of their capability for solving their problems.

- Proposing the bottom-up approach as an alternative to the top-down approach for dissemination.

The traditional top-down approach for dissemination maximizes internal validity; initial evaluations in the top-down approach are efficacy evaluations that apply RCTs. After internal validity is established, to achieve high external validity, evaluators carry out subsequent effectiveness evaluations on the same intervention in real-world settings. The top-down sequential approach is traditionally regarded by many researchers as the scientific path to validity achievement. This strategy has been widely used in biomedical research with fruitful results.

However, this dissemination approach does not work so well in social betterment/health promotion programs. As discussed previously, efficacious interventions are not necessarily effective in the real world, or do not address all issues of stakeholders' interest. This leads to a huge gap between the academic and the practice communities regarding interventions. (Glasgow et al., 2006; Glasgow, Lichtenstein, & Marcus, 2003; Green & Glasgow, 2006; Wandersman, 2003; Wandersman et al., 2008). To address the problem, social betterment/health promotion programs require an alternative approach. Pragmatic synthesis proposes the use of the bottom-up approach as an alternative route from evaluation to dissemination (Chen, 2010; Chen & Garbe, 2012). This sequential approach occurs in the reverse order of the top-down approach. To maximize viability (i.e., is the intervention practical, affordable, suitable, and helpful), the evaluation starts with viability evaluation. If the real-world intervention is found to be viable, an *effectiveness evaluation* is conducted next to provide systematic and objective evidence of the intervention's effectiveness in the real world. If necessary, the effectiveness evaluation could also address issues on whether the effectiveness is generalizable to other real world settings. After the intervention is found to be viable, effective, and generalizable in real world evaluations, an *efficacy evaluation* using methods such as RCTs are conducted to rigorously assess the causal relationship between the intervention and the outcome.

Fig. 2 highlights the differences between the bottom-up and top-down approaches:

The bottom-up approach could have the following potential merits: assuring the intervention's usefulness to stakeholders, avoiding wasting money, and providing an opportunity to revise and improve the intervention in a real world setting before finalizing its protocol (Chen & Garbe, 2012).

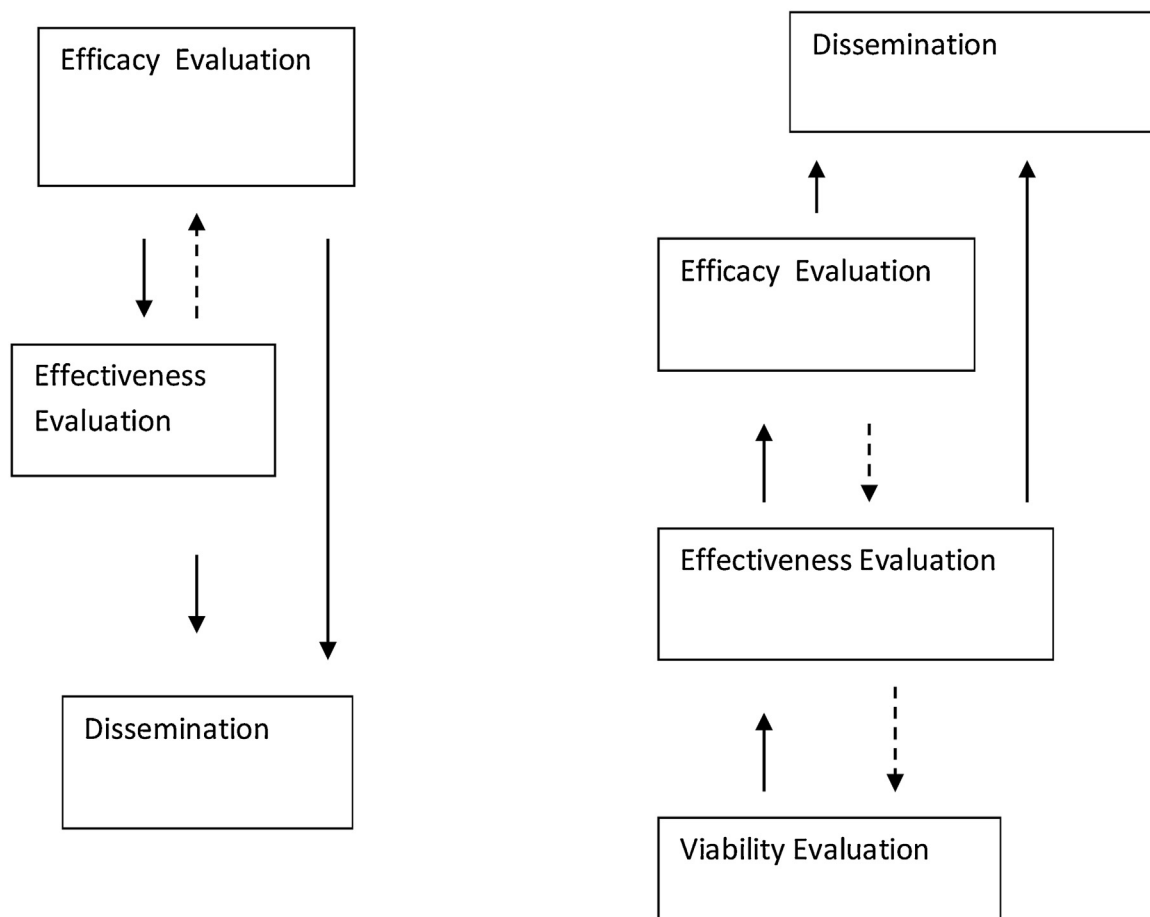
4.3. Advantages and challenges for applying pragmatic synthesis to evaluation

4.3.1. Advantages

- Provides a theoretical perspective for guiding evaluation of moderate-complexity or real-world programs.

Evaluators working with moderate-complexity programs have been relied on reductionism or systems thinking for guiding evaluations. However, these theories may not appropriately cover the range of situations especially important to moderate-complexity programs. With the availability of pragmatic synthesis, evaluators have a better perspective to follow for assessing these program and better serving stakeholders' evaluation interest and needs.

Top-Down Approach



Bottom-Up Approach

Fig. 2. The top-down approach vs. bottom-up approach.

- Incorporating ideas from systems thinking without the information overload or overwhelming complexity.

As discussed previously, evaluators interested in applying systems thinking may be hindered by its information overload. Pragmatic synthesis provides an option for evaluators to address some issues raised by systems thinking without being intimidated by its complexity by assuming that *everything* relates to *everything else* in the system as argued by some advocates of systems thinking. For example, concepts and approaches proposed by pragmatic synthesis such as adjuvants, joint effects, viability

evaluation, and the bottom-up approach provide more manageable and applicable options for addressing systems issues.

- Learning from community

In disciplines such as medicine, researchers hold the best knowledge and technologies available for solving health problems. In solving social problems, responsibility for knowledge and technologies rests with both, researchers and stakeholders. Researchers have plenty of platforms for cumulating their knowledge and technology, but few are available for learning

from communities. Pragmatic synthesis has a potential for providing a platform for formally testing and cumulating stakeholders' knowledge and technologies.

- Opening a window of opportunity for developing cutting-edge concepts and approaches to conduct evaluation.

Pragmatic synthesis attempts to reconcile the differences between reductionism and systems thinking. As illustrated in this article, the reconciliation could generate new ideas and insights for developing cutting-edge evaluation concepts and approaches.

4.3.2. Challenges

- Gaining evaluators' recognition and acceptance

As a new perspective with a short history, pragmatic synthesis does not have the critical mass supporters, reputation and prestige as reductionism and systems thinking have been enjoying. The perspective must attract evaluators to join and further contribute to develop this theory and its applications in order to grow. This perspective is at the beginning of its development and much work needs to be done. It is encouraging to see evaluators applying this perspective to their practice (Flygare, Gill, & Johansson, 2013; Hashima, 2014; Hawk, 2014). An evaluation of a colon cancer screening program using this perspective as well is currently undergoing.

- Acquiring funding agencies' support for investigating and documenting the pragmatic synthesis merit and contribution to improving the quality and thoroughness of evaluation.

Another challenge faced by pragmatic synthesis is to acquire support from funding agencies and foundations. The expansion and dissemination of an evaluation perspective or approach would be much easier with support from funding agencies. For example, NIH supports projects which apply RCTs. The support sustains the prestige of RCTs and reductionism. CDC requires grantees to apply logic models. The requirement enhances the popularity of logic models. Finding an agency or foundation to support pragmatic synthesis and its approaches will be a top priority for promoting pragmatic synthesis in the future.

5. Implications for future directions

The above discussions of the theoretical perspectives and approaches raise the following crucial issues that need to be addressed for further advancing program evaluation in the future.

5.1. How can evaluators select an appropriate perspective or approach?

With a variety of theories and approaches, we need strategies to provide evaluators guidance on selecting an appropriate theoretical perspective or approach for their evaluations. Two general strategies are available for providing such guidance: universalist and contingency.

5.1.1. Universalist strategy

This view is based upon the belief that there is a perspective or approach universally superior to others, implying the existence of a ranking from best to least fit approach. This strategy has been widely applied in industry and medicine. For example, Consumer Reports provide annual ranking of the *best product* including automobile, television, washing machine, etc to guide consumers' purchasing needs. The universalist strategy has influenced evaluation thinking in a similar way. For example, some evaluators

argue that there is a best method (e.g., RCTs), a best intervention (e.g. evidence-based interventions), best practice etc. This strategy also stimulated debates in the evaluation community on whether quantitative or qualitative methods are the best. While the universalist strategy provides a clear, straightforward guide for selecting an evaluation approach based on the *best fit* principle, it is also conducive of an *one-size-fits-all* type of thinking in evaluation practice.

5.1.2. Contingency strategy

The contingency strategy proposes that there is no such thing as the best perspective, approach, or method in evaluation. Instead, *different situations require different perspectives* or approaches. The selection depends on factors such as the nature of the program to be evaluated, the purpose of evaluation, and the stakeholders' needs. For example, the complexity of the program bears major weight on selecting a theoretical evaluation perspective. Thus, for simple or stable programs, reductionism is highly suitable, highly complex programs may require using systems thinking, while real-world programs of moderate-complexity may better benefit from applying pragmatic synthesis. The contingency strategy, does not provide a clear-cut guide as the universalist strategy does; it builds however, a flexible problem-solving model, by emphasizing alternative ways to capture intervention effects most likely to be encountered in real life, therefore most relevant and useful to stakeholders. The author highly recommends this strategy for selecting a theoretical evaluation approach.

5.2. Should transdisciplinary theories be balanced by grass-root evaluation theories?

Evaluation has long benefited from adopting theories and approaches from matured sciences.

Their merits were already tested and documented in the literature, and they are readily available for evaluators to use. Many of the "imported" theories and approaches have contributed towards establishing evaluation as a discipline. However, it is important to recognize that these transdisciplinary theories and approaches were not developed to serve all evaluation needs, and cannot set the vision and agenda for future evaluation development. Grass-root theories and approaches must be developed to cover different aspects of conducting evaluation, with respect to satisfying not only a need for a guiding theoretical structure, but also a frame for accommodating local program and stakeholders' requirements. It is true that developing grass-root theories and approaches is highly challenging, and requires long time to build recognition and prestige, as opposed to using readily available theories from other disciplines. However, this is a path that we must take for delineating program evaluation as a strong and unique discipline, and to advance program evaluation as a mature applied science in the future.

6. Conclusions

This article discussed theories of program, theories of evaluation, and their relationships under three theoretical perspectives: Reductionism, systems thinking, and pragmatic synthesis. Each theoretical perspective has its own theories of program and theories of evaluation to guide and support evaluation practice. An understanding of these perspectives, their relative strengths and challenges, and related issues are helping evaluators to choose appropriate evaluation designs and better service stakeholders' evaluation needs. This article demonstrates that knowledge in these areas can provide a better guide for designing fruitful evaluations, improving quality of evaluation practice, informing potential areas for developing cutting-edge evaluation approaches,

and contributing in advancing program evaluation toward a matured applied science.

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